

PROJECT STAGE - I

on

**An Enhanced Deep Learning Technique for Detection
of Acute Pneumonia**

Submitted in partial fulfillment of the requirements for the award of a degree of

BACHELOR OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING

by

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College of Engineering for Women

(Approved by AICTE, New Delhi and Affiliated to JNTUH, Hyderabad)

Accredited by NBA and NAAC with A Grade

Bachupally, Hyderabad – 500090

December, 2023

DECLARATION

We hereby declare that the work described in this report, entitled “**An Enhanced Deep Learning Technique for Detection of Acute Pneumonia**” which is submitted by us in partial fulfillment for the award of the degree of **Bachelor of Technology** in the Department of **Computer Science And Engineering** at **BVRIT HYDERABAD College of Engineering for Women**, affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500085 is the result of original work carried out under the guidance of **Ms. M. Shanmuga Sundari, Assistant Professor, Department of CSE.**

This work has not been submitted for any Degree / Diploma of this or any other institute/university to the best of our knowledge and belief.

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CERTIFICATE

This is to certify that the project work report, entitled “**An Enhanced Deep Learning Technique for Detection of Pneumonia**” is a bonafide work carried out by **Ms. Tinnaluri Yasaswini (20WH1A0518), Ms. Chandana Sai Bodicherla (20WH1A0522), Ms. Tella Tharunika (20WH1A0542)** in partial fulfillment for the award of B. Tech degree in **Computer Science and Engineering, BVRIT HYDERABAD College of Engineering for Women, Bachupally, Hyderabad**, affiliated to **Jawaharlal Nehru Technological University Hyderabad**, under my guidance and supervision.

The results embodied in the project work have not been submitted to any other university or institute for the award of any degree or diploma.

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ABSTRACT

Pneumonia is a potentially life-threatening acute lung infection, that manifests with varying severity levels, ranging from mild to critical. Swift and accurate assessment of its severity is crucial for timely and appropriate treatment, significantly impacting patient outcomes. Traditionally, chest X-rays serve as the primary imaging method for pneumonia diagnosis and severity evaluation. However, manual interpretation of these X-rays can be subjective and time-intensive. Leveraging the power of deep learning, particularly Convolutional Neural Networks (CNNs), offers a promising solution. In this context, we propose the utilization of a sophisticated deep learning model, specifically the DenseNet-121 architecture, for automated pneumonia severity assessment on chest X-ray images. Trained on an extensive dataset of chest X-rays, our model demonstrates the potential to enhance the precision and efficiency of pneumonia severity evaluation. By introducing automation into this critical diagnostic process, our approach is not a significant stride towards more effective and cost-efficient healthcare practices.

TABLE OF CONTENTS

S. No	Topic	Page no
1.	Introduction	8
	1.1 Problem Statement	
	1.2 Objective	
	1.3 Existing System	
	1.4 Proposed System	
2.	Literature Survey	10
3.	Requirements	14
	3.1 Software Requirements	
	3.2 Hardware Requirements	
4.	Architecture	15
5.	Methodology	16
	5.1 DenseNet121	
6.	Implementation and Results	17
	6.1 Code	
	6.2 Results	
7.	Extension Plan (For Phase – II)	22
8.	Dataset and References	23

LIST OF FIGURES

S.no	Description	Page.no
1.	The architecture of pneumonia detection	15
2.	Layers of DenseNet121 architecture	16
3.	Confusion Matrix	21
4.	ROC Curve	21
5.	a) Normal 1 b) Normal 2 c) Normal 3	25
6.	a) Pneumonia 1 b) Pneumonia 2 c) Pneumonia 3	25

1. INTRODUCTION

1.1 Problem statement:

This model involves the usage of DenseNet121, a prominent convolutional neural network, for the precise detection of pneumonia from chest X-ray images. Leveraging its deep and symmetric architecture, DenseNet121 demonstrates promising capabilities in distinguishing between normal and pneumonia-affected cases.

1.2 Objective:

The primary objective of this project is to evaluate the effectiveness of the DenseNet121 convolutional neural network in accurately detecting pneumonia from chest X-ray images. By delving into advanced deep learning techniques, the project aims to contribute novel insights for refining diagnostic approaches in respiratory health. Ultimately, the study aspires to elevate the accuracy and efficiency of pneumonia detection through the utilization of the DenseNet121 architecture.

1.3 Existing system:

The existing system, built upon the DenseNet121 convolutional neural network, represents a robust framework for pneumonia detection in chest X-ray images. The utilization of DenseNet121 enhances the efficiency and reliability of the current system, showcasing its efficacy as a valuable tool for automated and precise pneumonia diagnosis.

1.4 Proposed system:

1.4.1 Dataset Description and Methodology:

The dataset employed in this study comprises 5,856 chest X-ray images categorized into two classes. Pneumonia and Normal are the classes in the dataset. This dataset provides a diverse representation of pulmonary conditions, essential for training and evaluating the model's performance. The balanced distribution of images between the two classes ensures a comprehensive assessment of the model differentiation abilities.

The chosen methodology involves implementing the DenseNet121 architecture for pneumonia detection. DenseNet121, recognized for its dense connectivity and efficient feature reuse, offers a robust foundation for accurate image classification tasks. The model will undergo a rigorous training process on the dataset, leveraging its intricate architecture to learn and capture intricate patterns indicative of pneumonia. Subsequently, the trained DenseNet121 model will be evaluated on separate test data to gauge its efficacy in distinguishing between Pneumonia and Normal cases, thereby validating its suitability for automated pneumonia diagnosis.

2. LITERATURE SURVEY

Title: A Novel Method for Multivariate Pneumonia Classification Based on Hybrid CNN-PCA Based Feature Extraction Using Extreme Learning Machine with CXR Images.

The research paper titled "A Novel Method for Multivariate Pneumonia Classification Based on Hybrid CNN-PCA Based Feature Extraction Using Extreme Learning Machine with CXR Images" by MD. Nahiduzzaman, Md. Omar Faruq Goni and Saravanakumar Gurusamy introduce a sophisticated hybrid approach for pneumonia classification utilizing chest X-ray (CXR) images. The method combines Convolutional Neural Networks (CNNs) with Principal Component Analysis (PCA)-based feature extraction and Extreme Learning Machines (ELM), presenting a novel and potentially effective solution for multivariate pneumonia classification.

However, the paper lacks specific details regarding the models used, accuracy results, and datasets employed. To comprehensively evaluate the proposed method's effectiveness, future work should provide in-depth information on the CNN architectures implemented, the outcomes of the PCA-based feature extraction, and the performance metrics, including accuracy rates. Additionally, a thorough description of the datasets used, including size, diversity, and any preprocessing steps, would enhance the paper's credibility and allow for a more nuanced understanding of the proposed hybrid approach. Providing these details would contribute to the transparency and reproducibility of the research findings.

Title: An Interpretable Deep Learning Model for Covid-19 Detection with Chest X-Ray Images

The research paper titled "An Interpretable Deep Learning Model for Covid-19 Detection with Chest X-Ray Images" authored by Gurmail Singh and Kin-Choong Yow presents a significant contribution to the field by focusing on the development of an interpretable deep-learning model for COVID-19 detection using chest X-ray images. The emphasis on interpretability is crucial in the medical domain, as it not only facilitates trust in the model's predictions but also aids healthcare professionals in understanding the reasoning behind the outcomes. However, to comprehensively assess the impact of the proposed model, additional details regarding the

dataset and the specific deep learning architecture used are essential. Information about the dataset, such as size, diversity, and any preprocessing steps, is critical to understanding the generalizability of the model. Furthermore, insights into the chosen deep learning architecture, its complexity, and any specific interpretability techniques employed would provide valuable context for the reader and the wider scientific community.

In future work, the authors could enhance the paper's completeness by including these essential details, enabling a more thorough evaluation of the proposed interpretable deep learning model for COVID-19 detection. This would contribute to the transparency, reproducibility, and applicability of the research findings in real-world clinical settings.

Title: Automatic Detection of COVID-19 Infection Using Chest X-Ray Images Through Transfer Learning

The research paper titled "Automatic Detection of COVID-19 Infection Using Chest X-Ray Images Through Transfer Learning" by Elene Firmeza Ohata and Gabriel Maia Bezerra introduces a promising avenue for combating the COVID-19 pandemic by leveraging transfer learning for chest X-ray image analysis. The authors rightly highlight the potential of this approach to achieve high accuracy, emphasizing the significance of early detection in the effective management of COVID-19 patients. The paper contributes to the growing body of literature exploring advanced techniques in medical imaging for rapid and accurate diagnosis, particularly in the context of infectious diseases.

However, to comprehensively assess the impact of the proposed transfer learning approach, additional details are needed. Specifically, the paper lacks information about the specific transfer learning model employed, accuracy results, and details regarding the dataset used in the research. Without these crucial details, it is challenging to thoroughly evaluate the effectiveness and generalizability of the model. Future work should address this gap by providing comprehensive insights into the transfer learning architecture, reporting accuracy metrics, and offering a detailed description of the dataset, enabling researchers and practitioners to better understand, replicate, and build upon the study's findings.

Title: Pneumonia Detection Proposing a Hybrid Deep Convolutional Neural Network Based on Two Parallel Visual Geometry Group Architectures and Machine Learning Classifiers.

The research paper titled "Pneumonia Detection: Proposing a Hybrid Deep Convolutional Neural Network Based on Two Parallel Visual Geometry Group Architectures and Machine Learning Classifiers" by Mohammad Yaseliani and Ali Zeal Hamada addresses a critical aspect of medical diagnostics by emphasizing the importance of pneumonia detection. The focus on developing a hybrid

An Enhanced Deep Learning Technique for Detection of Acute Pneumonia

model integrating two parallel Visual Geometry Group (VGG) architectures and machine learning classifiers suggests a nuanced approach to improving accuracy in pneumonia identification.

Given the significance of early detection in pneumonia cases, the proposed hybrid model holds the potential for enhancing diagnostic capabilities and subsequently facilitating timely medical intervention, ultimately contributing to improved patient outcomes. The paper aligns with the broader trend of leveraging advanced technologies to address crucial challenges in the medical field, highlighting the potential of hybrid deep-learning approaches for pneumonia detection.

However, the paper appropriately acknowledges the potential challenges associated with hybrid models, particularly in terms of computational resources. The complexity of such models may pose implementation challenges, especially in resource-constrained environments. This recognition of potential limitations is important for guiding future research and applications, prompting consideration for optimization strategies or alternative approaches that balance model performance with practical implementation feasibility. As part of future work, exploring ways to streamline or optimize the proposed hybrid model for more efficient use of computational resources could enhance the model's applicability in a broader range of medical settings.

Title: Convolutional Sparse Support Estimator-Based COVID-19 Recognition From X-Ray Images

The research conducted by Mehmet Yamaç and Mete Ahishali focuses on COVID-19 recognition from X-ray images using a technique known as "Convolutional Sparse Support Estimator" (CSEN). The mention of CSEN suggests the application of advanced techniques for feature extraction and pattern recognition, indicating a potentially sophisticated approach to accurate COVID-19 recognition.

However, the provided information lacks specific details about the CSEN-based method employed, accuracy results, and the dataset utilized in the research. The absence of these crucial details makes it challenging to comprehensively evaluate the effectiveness of the model. To gain a thorough understanding of the research's outcomes and contributions, additional information on the methodology, accuracy metrics, and dataset characteristics would be essential.

Title: Deep GRU-CNN Model for COVID-19 Detection From Chest X-Rays Data

The collaborative work by Abdullah Gani, Carsten Maple, and Yulin Wang introduces a Deep GRU-CNN Model for COVID-19 detection from chest X-ray data. The integration of Gated Recurrent Units (GRU) and Convolutional Neural Networks (CNN) in the model implies a deep learning architecture designed to

capture both spatial and temporal features within chest X-ray data. This combination suggests a sophisticated approach that holds the potential for accurate COVID-19 detection.

However, the provided information lacks specific details about the deep GRU-CNN model employed, accuracy results, and the dataset utilized in the research. The absence of these crucial details poses a challenge in comprehensively evaluating the effectiveness of the model. To thoroughly assess the contributions and outcomes of the research, additional information regarding the model architecture, performance metrics, and dataset characteristics would be essential.

Title: Deep Learning Algorithms for Automatic COVID-19 Detection on Chest X-Ray Images

The research conducted by Sergio Cannata and Annunziata Paviglianiti centers around the application of deep learning algorithms for the automatic detection of COVID-19 in chest X-ray images. The focus on deep learning algorithms implies an exploration of advanced techniques for COVID-19 detection, particularly within the realm of image-based medical diagnosis, where deep learning models have demonstrated considerable promise.

However, the provided information lacks specific details regarding the deep learning algorithms utilized, accuracy results, and the dataset employed in the research. The absence of these crucial details makes it challenging to comprehensively evaluate the effectiveness of the models and the overall impact of the research. To gain a thorough understanding of the outcomes and contributions of the study, additional information on the employed algorithms, performance metrics, and dataset characteristics would be essential.

3. REQUIREMENTS

3.1 Software Requirements

- **Language Used:** Python 3.8
- **Operating System:** Windows
- Google Colab

3.2 Hardware Requirements

- **Processor:** Intel Core i5/i7 processor
- **RAM:** 8GB

4. ARCHITECTURE

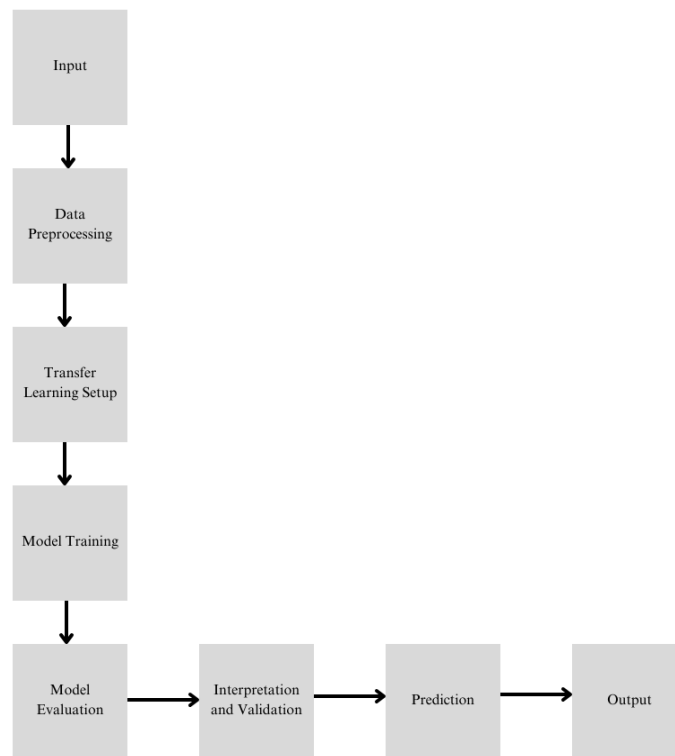


Fig 1. The architecture of pneumonia detection

5. METHODOLOGY

5.1 DENSENET121:

DenseNet121 is a convolutional neural network (CNN) architecture that belongs to the DenseNet family. The key idea behind DenseNet is to connect each layer to every other layer in a dense block, promoting feature reuse.

The architecture of DenseNet121:

1. **Dense Blocks:** DenseNet introduces the concept of dense blocks, where each layer is connected to every other layer in a feedforward fashion. This connectivity pattern enhances feature reuse, allowing the network to efficiently learn representations. Dense blocks are composed of densely connected convolutional layers.
2. **Transition Blocks:** Between dense blocks, transition blocks are used to reduce the spatial dimensions (width and height) of the feature maps. This is achieved through a combination of a 1×1 convolutional layer and average pooling.
3. **Batch Normalization and ReLU Activation:** Each convolutional layer is followed by batch normalization and a rectified linear unit (ReLU) activation function. Batch normalization helps stabilize and accelerate training, while ReLU introduces non-linearity to the network.
4. **Global Average Pooling (GAP):** Instead of using fully connected layers at the end of the network, DenseNet employs global average pooling. This spatial pooling operation computes the average value for each feature map, resulting in a fixed-size vector. GAP helps reduce overfitting and decreases the number of parameters.
5. **SoftMax Classifier:** The output of the global average pooling layer is connected to a SoftMax classifier, which produces the final class probabilities.

Layers	Output Size	DenseNet-121	DenseNet-169	DenseNet-201	DenseNet-264
Convolution	112×112	7×7 conv, stride 2			
Pooling	56×56	3×3 max pool, stride 2			
Dense Block (1)	56×56	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 6$
Transition Layer (1)	56×56	1×1 conv			
	28×28	2×2 average pool, stride 2			
Dense Block (2)	28×28	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 12$
Transition Layer (2)	28×28	1×1 conv			
	14×14	2×2 average pool, stride 2			
Dense Block (3)	14×14	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 24$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 48$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 64$
Transition Layer (3)	14×14	1×1 conv			
	7×7	2×2 average pool, stride 2			
Dense Block (4)	7×7	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 16$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 32$	$\begin{bmatrix} 1 \times 1 \text{ conv} \\ 3 \times 3 \text{ conv} \end{bmatrix} \times 48$
Classification Layer	1×1	7×7 global average pool			
		1000D fully-connected, softmax			

Fig 2. Layers of DENSENET121 architecture

6. IMPLEMENTATION AND RESULTS

6.1 Code

```
from google.colab import drive
drive.mount('/content/drive')
```

```
import tensorflow as tf
from tensorflow.keras.models import Model
from tensorflow.keras.layers import Dense, GlobalAveragePooling2D
from tensorflow.keras.applications import DenseNet121
from tensorflow.keras.optimizers import Adam
from tensorflow.keras.preprocessing.image import ImageDataGenerator
from tensorflow.keras.callbacks import LearningRateScheduler
```

```
# Constants
IMAGE_SIZE = (224, 224)
BATCH_SIZE = 32
NUM_CLASSES = 2 # Normal and Pneumonia
```

```
# Data Generators with Data Augmentation
```

```
train_datagen = ImageDataGenerator(
    rescale=1./255,
    rotation_range=20,
    width_shift_range=0.2,
    height_shift_range=0.2,
    shear_range=0.2,
    zoom_range=0.2,
    horizontal_flip=True,
    fill_mode='nearest'
)
```

```
train_generator = train_datagen.flow_from_directory(
    '/content/drive/MyDrive/Major_Project_Pneumonia_Detection/D2/train'
,
    target_size=IMAGE_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='categorical'
)
```

```
test_datagen = ImageDataGenerator(rescale=1./255)
test_generator = test_datagen.flow_from_directory(
    '/content/drive/MyDrive/Major_Project_Pneumonia_Detection/D2/test',
    target_size=IMAGE_SIZE,
```

An Enhanced Deep Learning Technique for Detection of Acute Pneumonia

```
batch_size=BATCH_SIZE,
class_mode='categorical',
shuffle=False
)

validation_datagen = ImageDataGenerator(rescale=1./255)
validation_generator = validation_datagen.flow_from_directory(
    '/content/drive/MyDrive/Major_Project_Pneumonia_Detection/D2/val',
    target_size=IMAGE_SIZE,
    batch_size=BATCH_SIZE,
    class_mode='categorical'
)

from tensorflow.keras.layers import Dropout

# Load Pretrained DenseNet-169 Model
base_model = DenseNet121(weights='imagenet', include_top=False,
input_shape=(224, 224, 3))

# Add Custom Layers for Classification
x = base_model.output
x = GlobalAveragePooling2D()(x)

x = Dropout(0.5)(x)
x = Dense(512, activation='relu')(x)

x = Dropout(0.2)(x)
x = Dense(256, activation='relu')(x)

x = Dropout(0.2)(x)
x = Dense(128, activation='relu')(x)

x = Dropout(0.2)(x)
x = Dense(64, activation='relu')(x)

predictions = Dense(NUM_CLASSES, activation='softmax')(x)

model = Model(inputs=base_model.input, outputs=predictions)

# Compile the Model
model.compile(optimizer=Adam(learning_rate=0.0001),
loss='categorical_crossentropy', metrics=['accuracy'])

# Calculate steps_per_epoch and validation_steps
steps_per_epoch = len(train_generator)
validation_steps = len(validation_generator)
```

An Enhanced Deep Learning Technique for Detection of Acute Pneumonia

```
# Learning Rate Scheduler
def lr_schedule(epoch):
    initial_learning_rate = 0.0001
    decay = 0.9
    if epoch % 2 == 0:
        return initial_learning_rate * decay
    else:
        return initial_learning_rate

lr_callback = LearningRateScheduler(lr_schedule)

# Train the Model with Correct steps_per_epoch and validation_steps
model.fit(
    train_generator,
    steps_per_epoch=steps_per_epoch,
    epochs=10,
    validation_data=validation_generator,
    validation_steps=validation_steps,
    callbacks=[lr_callback]
)

# Save the Model
model.save('/content/drive/MyDrive/Major_Project_Pneumonia_Detection/models/pneumonia_densenet121.h5')

from tensorflow.keras.preprocessing import image
from tensorflow.keras.models import load_model
import numpy as np

# Load the pre-trained DenseNet-121 model
model =
load_model('/content/drive/MyDrive/Major_Project_Pneumonia_Detection/models/pneumonia_densenet121.h5')

# Function to preprocess the user input image
def preprocess_image(image_path):
    img = image.load_img(image_path, target_size=(224, 224))
    img_array = image.img_to_array(img)
    img_array = np.expand_dims(img_array, axis=0)
    img_array /= 255.0 # Normalize pixel values
    return img_array

# Take input image path from the user
input_image_path = input("Enter the path to the image you want to
classify: ")

try:
```

An Enhanced Deep Learning Technique for Detection of Acute Pneumonia

```
# Preprocess the input image
input_img_array = preprocess_image(input_image_path)

# Perform prediction
predictions = model.predict(input_img_array)
class_labels = ['Normal', 'Pneumonia']
predicted_class_index = np.argmax(predictions)
predicted_class = class_labels[predicted_class_index]
confidence = predictions[0][predicted_class_index] * 100

# Display the prediction
print(f"The input image is classified as {predicted_class} with
confidence {confidence:.2f}%.")

except Exception as e:
    print(f"Error: {e}")
```

```
from sklearn.metrics import accuracy_score, confusion_matrix,
classification_report
import matplotlib.pyplot as plt
import seaborn as sns

# Evaluate the model using the test generator
test_loss, test_accuracy = model.evaluate(test_generator,
steps=len(test_generator))
print(f'Test Accuracy: {test_accuracy:.2f}')
print(f'Test Loss: {test_loss:.2f}')

# Predict using the model
predictions = model.predict(test_generator)
predicted_labels = np.argmax(predictions, axis=1)

# True labels
true_labels = test_generator.classes

# Calculate accuracy
accuracy = accuracy_score(true_labels, predicted_labels)
print(f'Overall Accuracy: {accuracy:.2f}')

# Generate and print classification report
report = classification_report(true_labels, predicted_labels,
target_names=['Normal', 'Pneumonia'])
print(f'Classification Report:\n{report}')

# Generate and print confusion matrix
matrix = confusion_matrix(true_labels, predicted_labels)
```

An Enhanced Deep Learning Technique for Detection of Acute Pneumonia

```
print(f'Confusion Matrix:\n{matrix}')

# Plot the confusion matrix
plt.figure(figsize=(6, 6))
sns.heatmap(matrix, annot=True, fmt='g', cmap='Blues',
            xticklabels=['Normal', 'Pneumonia'],
            yticklabels=['Normal', 'Pneumonia'])
plt.xlabel('Predicted Labels')
plt.ylabel('True Labels')
plt.title('Confusion Matrix')
plt.show()

# ROC Curve
from sklearn.metrics import roc_curve, auc

fpr, tpr, _ = roc_curve(true_labels, predictions[:, 1]) # Assuming
predictions[:, 1] gives probabilities for the positive class
roc_auc = auc(fpr, tpr)

plt.figure(figsize=(8, 6))
plt.plot(fpr, tpr, color='darkorange', lw=2, label=f'ROC curve (area =
{roc_auc:.2f})')
plt.plot([0, 1], [0, 1], color='navy', lw=2, linestyle='--')
plt.xlabel('False Positive Rate')
plt.ylabel('True Positive Rate')
plt.title('ROC Curve')
plt.legend(loc='lower right')
plt.show()
```

6.2 Results

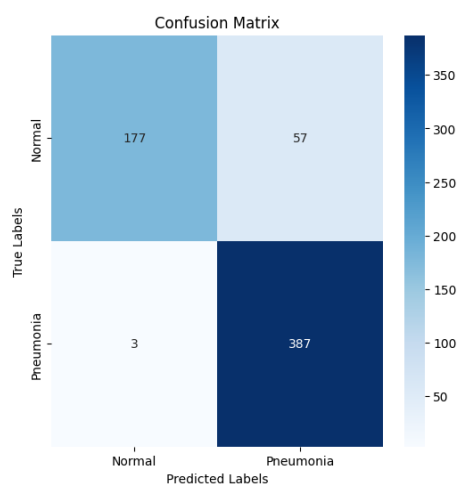


Fig 3. Confusion Matrix

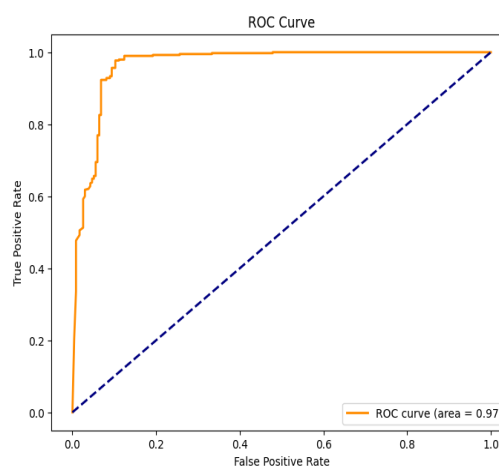


Fig 4. ROC Curve

7. EXTENSION PLAN (FOR PHASE - II)

- Detecting Pneumonia and Normal chest X-ray.
- Implementing to markup the affected region of the Pneumonia in the chest x-ray

8. DATASET AND REFERENCES

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Sample Dataset

- Fig 5. Normal Chest X-ray images

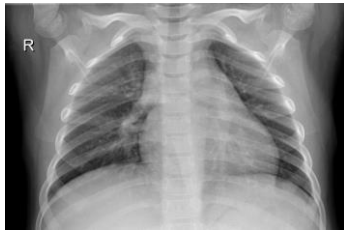


Fig 5a. Normal 1



Fig 5b. Normal 2



Fig 5b. Normal 3

- Fig 6. Pneumonia Chest X-ray images



Fig 6a. Pneumonia 1

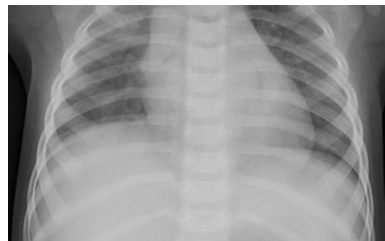


Fig 6b. Pneumonia 2



Fig 6c. Pneumonia 3